Regd. No.

VIGNAN'S

(Deemed to be University)

Regulation: R22 **Code: 22EC201/1**

II B. Tech. I Semester Regular Examinations –January, 2023

ANALOG CIRCUITS

(ECE)

Time: 150 Min **Max. Marks: 80M**

SECTION – A

Answer all Four questions **4×10M=40M**

1. A microphone having an output resistance of 1 kΩ generates a peak signal level of 2 mV. Design a CE stage with a bias current of 1 mA that amplifies this signal to 40 mV. Assume $R_E = 4/g_m$ and $\beta=100$.

2. Consider the two inverting amplifiers in cascade in Figure 1. The op-amp parameters are $A_{OL} = 5 \times 10^3$, $R_i = 10$ kΩ, and $R_o = 1$ kΩ. Determine the actual closed-loop gains $A_{vf1} = v_{o1}/v_i$ and $A_{vf} = v_{o2}/v_i$. What is the percent error from the ideal values?

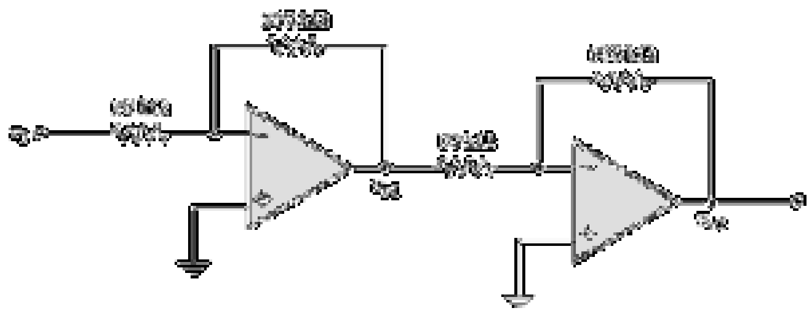

Figure 1

3. In the inverted-ladder DAC shown the switches are connected directly to the Op-Amp input.

   (a) Show that the current ***I*** drawn from V_R is a constant independent of the digital word.

   (b) What is the switch current and V_o if the MSB is 1 and all other bits are zero?

   (c) Repeat (b), assuming that the next MSB is 1 and all other bits are zero.

   (d) Calculate V_o for the LSB in the 4-bit D/A converter with all other bits zero.

4. In the design of a particular variable-frequency Wien bridge oscillator, the values of R_1 and C_2 are adjustable according to the relations $0.1 \le R_1/R_2 \le 10$ and $0.1 \le C_2/C_1 \le 10$.

   (a) Find the minimum and maximum values of the oscillation frequency for $R_2 = 10$ kΩ and $C_1=0.1$ uF.

   (b) What must be the minimum gain-bandwidth product of the voltage amplifier if the amplifier frequency response is to have a negligible effect on oscillator performance?

SECTION-B

Answer all Two questions **2×20M=40M**

5. A two-stage audio voltage preamplifier is shown in Figure 2. The first stage is a common-emitter pnp with voltage-divider bias, and the second stage is a common-emitter npn with voltage-divider bias. It has been decided that the amplifier should operate from 30 V dc to get a large enough signal voltage swing to provide a maximum of 6 W to the speaker. Because small IC regulators such as the 78xx and 79xx series are not available above 24 V, dual {15 V dc supplies are used in this particular system instead of a single supply. The operation is essentially the same as if a single +30 V dc source had been used. The potentiometer at the output provides gain adjustment for volume control. The input to the first stage is from the microphone, and the output of the second stage will drive a power amplifier. The power amplifier will drive the speaker. The preamp is to operate with a peak input signal range of from 25 mV to 50 mV. The minimum range of voltage gain adjustment is from 90 to 170.

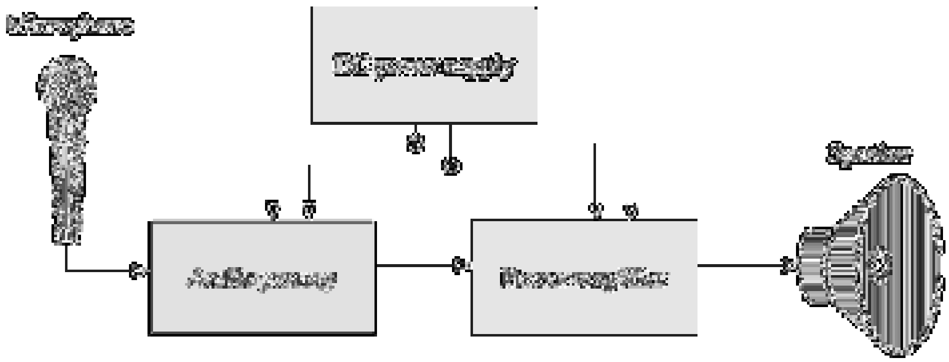

Figure 2

6. Design an astable multivibrator with f_0=4 KHz and D (%) = 55% by using 555 timer. The timing accuracy of the 555 astable approaches 1%, with a temperature stability of 0.005%/ºC and a power supply stability of 0.05%/V.

   a) Since TH>TL, the circuit always gives D (%) > 50%. Find a better approach to get perfect symmetry.

   b) Assuming Vcc=6V in the multivibrator, find the range of variation of f_0 and D (%) if the voltage at the control input is modulated by ac coupling to it external sine wave with a peak amplitude of 1V.

   c) The node in the 555 timer at which the voltage is V_{TH} (i.e. the inverting input terminal of comparator 1) is usually connected to an external terminal. This allows the user to change V_{TH} externally (i.e. V_{TH} no longer remains at $\frac{2}{3}V_{CC}$). Note, however that whatever the value of V_{TH} becomes, V_{TL} always remains at $\frac{1}{2}V_{TH}$. For the astable multivibrator, rederive the expressions for T_H and T_L, expressing them in terms of V_{TH} and V_{TL}.